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1 Problems

The Numbers on a Blackboard: The numbers $1, 2, \dots, n$ are written on a blackboard. We may erase any two numbers a and b and write the number $a + b - 1$ instead. What number will remain after $n - 1$ operations?

Solution: Let a number $\mathbf{n_k} \in \{1, 2, \dots, n\}$. Now, in accordance with the given information, the expression $t = a + b - 1$ is considered. Here, $t_{\min} = 2$ and $t_{\max} = 2n - 2$. Suppose, after replacing a and b , we perform the same step for $a + b - 1$ and some w in the same set. This would give us $a + b + w - 2$ and so on. Thus:

$$a, b \rightarrow a + b - 1$$

$$a + b - 1, w \rightarrow a + b + w - 2$$

$$a + b + w - 2, x \rightarrow a + b + w + x - 3$$

and so on. Also, there may be another step, when we use for some integers:

$$a + b - 1, w + x - 1 \rightarrow a + b + x + w - 3. \quad (1)$$

Thus, we can in general state that after every third step we will get a number like in eq(1). Hence, after $(n-1)$ steps, we have a number of the form

$$(1 + 2 + 3 + \dots + n) - (n - 1)$$

And, thus giving me this number would result in deliberate steps to eventually end of having

$$t_{\text{end}} = \frac{n^2 - n + 2}{2} \quad (2)$$

Hence, the answer to this problem.

The Vector Swop: Start with the set of vectors $\{(1, 0), (0, 1), (-1, 0), (0, -1)\}$. In one step, you can pass from a set of vectors to a new set by replacing any two vectors $\vec{u}$ and $\vec{v}$ with $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$. Can you ever return to the initial configuration?

Solution: Initially, let the vector in general be (x, y) , where x and y are respective variables which vary very specifically in this case.

The **initial conditions** are:

If $x \in \{1, -1\}$, then $y = 0$.
If $y \in \{1, -1\}$, then $x = 0$.

Consider a state function for this condition i.e. a magnitude function of vector.

$$M(x, y) = x^2 + y^2 = 1 \quad (3)$$

Then, assume to vectors (x,y) and (k,z), which according to the state function gives

$$x^2 + y^2 = 1 \quad (4)$$

$$k^2 + z^2 = 1 \quad (5)$$

And now, we can state a change in function, using the given **step** rule. Here:

$$M_{old} = x^2 + y^2 = 1 \text{ or } k^2 + z^2 = 1 \quad (6)$$

$$\begin{aligned} M_{new} &= (x + k)^2 + (y + z)^2 \text{ or } (x - k)^2 + (y - z)^2 \\ &= x^2 + y^2 + k^2 + z^2 + 2(xk + yz) \text{ or } x^2 + y^2 + k^2 + z^2 - 2(xk + yz) \\ &= 2 + 2(xk + yz) \text{ or } 2 - 2(xk + yz) \end{aligned} \quad (7)$$

Here, we have problem with the M function to become complicated. But from observation, I can state that the sum of two magnitude functions of two different cases can combine to form something simpler. Let the combination of two cases be M_{com} .

This M_{com} can be stated as the sum of the magnitude of vectors that are changing. From previous cases,

$$M_{com1} = 2 \quad (8)$$

$$M_{com2} = 4 \quad (9)$$

$$M_{com3} = 8 \quad (10)$$

$$M_{com4} = 16 \quad (11)$$

, Here, accoring to the count of step the function increases, like 2 raised to power the count of step. As, there is no upper bound to this monotonic energy function, this process goes on forever and it never resemble the initial configuration.